Grade 11 · Algebra II · Topic 6: Exponential and Logarithmic Functions · Lesson 6-7 — Geometric Sequences and Series

Classwork — Give it a go

Show every step. Name the first term, the common ratio and the number of terms before you use a formula.

Student name :	Date : .....

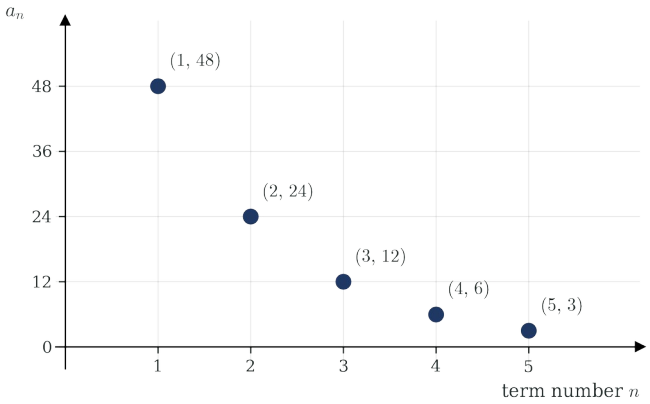
QUESTION 1 identify a geometric sequence

1	$7,\ 21,\ 63,\ 189,\ \ldots$	Show that the sequence is geometric. Find the common ratio and the next term.

QUESTION 2 not geometric

2	$2,\ 6,\ 12,\ 20,\ \ldots$	Show that this sequence is not geometric. Write every ratio you calculate.

QUESTION 3 from a graph



The graph shows the first five terms of a geometric sequence. Read the common ratio from the graph, write the explicit formula, and find the eighth term.

QUESTION 4 negative ratio

4	$243,\ -81,\ 27,\ \ldots$	Find the common ratio, then use the explicit formula to find the seventh term.

QUESTION 5 recursive and explicit

5	$5,\ 20,\ 80,\ \ldots$	Write the recursive formula and the explicit formula. Find the sixth term.

QUESTION 6 from two terms

6	$a_3 = 20, \quad a_6 = \frac{5}{2}$	The sequence is geometric. Show the number of steps between the terms, then find the common ratio and the first term.

QUESTION 7 from a recursive formula

7	$a_1 = 3, \quad a_n = -2\,a_{n-1}$	Write the explicit formula and find the eighth term.

QUESTION 8 sigma notation

8	$\sum_{k=1}^{5} 2 \cdot (-3)^{k-1}$	Write down the starting index, the ending index and the summand. Then evaluate by the formula and check by adding.

QUESTION 9 a ratio below one

9	$81 + 27 + 9 + \cdots$	Find the sum of the first 6 terms. Give the exact value as a fraction, then to 2 decimal places.

QUESTION 10 find the number of terms

10	$5 + 10 + 20 + \cdots$	The sum of the first n terms is 635. Find n, the number of terms.

QUESTION 11 model (model values only)

11	$a_1 = 2400, \quad r = 1.1$	A model Jeddah recycling centre sorts 2 400 kg of paper in week 1, and each week it sorts 1.1 times the weight of the week before. Find the weight sorted in week 8, and the total sorted in 8 weeks, each to the nearest kilogram.